

3.1.2	Amount of Substance	
	Relative atomic mass and relative molecular mass	be able to define relative atomic mass (A_r) and relative molecular mass (M_r) in terms of ^{12}C . (The term relative formula mass will be used for ionic compounds)
	The mole and the Avogadro constant (L)	understand the concept of a mole as applied to electrons, atoms, molecules, ions, formulae and equations
		understand the concept of the Avogadro constant. (Calculation not required)
	The ideal gas equation	be able to recall the ideal gas equation $pV = nRT$ and be able to apply it to simple calculations in S.I. units, for ideal gases
	Empirical and molecular formulae	understand the concept of, and the relationship between, empirical and molecular formulae
		be able to calculate empirical formulae from data giving percentage composition by mass
	Balanced equations and associated calculations	be able to write balanced equations (full and ionic) for reactions studied
		be able to balance equations for unfamiliar reactions when reactants and products are specified. (This is an important skill that applies in all units.)
		be able to calculate reacting volumes of gases
		be able to calculate concentrations and volumes for reactions in solutions, limited to titrations of monoprotic acids and bases and examples for which the equations are given
		know that $\% \text{ atom economy} = \frac{\text{mass of desired product}}{\text{total mass of reactants}} \times 100$
		be able to calculate reacting masses, % yields and % atom economies from balanced equations